# Vikram Singh

Embedded Software Engineer

## Contact

vikram.singh@example.com
+91 90000 00014
Bengaluru, India
github.com/vikram-example

## Skills

**Languages:** C, C++, Embedded C, Python

**Hardware:** ARM Cortex-M, STM32, ESP32

**Protocols:** UART, SPI, I2C, CAN bus

**Software:** FreeRTOS, RTOS, Embedded Linux, device drivers, MISRA C

**Tools:** GDB, JTAG, oscilloscope, logic analyzer, Git, Makefile, CMake

## Education

B.Tech in Electronics, Example Institute of Technology (2018 - 2022), CGPA 8.0

## Certifications

- Embedded Systems with ARM Cortex-M (online)
- Embedded Linux basics course

## Profile

Embedded engineer with 4 years of experience writing real-time firmware in C and C++ for ARM Cortex-M microcontrollers. Skilled at hardware debugging and driver development.

## Experience

### Embedded Software Engineer, Electra Controls

Jun 2023 - Present

- Wrote firmware in C and C++ for STM32 ARM Cortex-M microcontrollers used in motor controllers.
- Implemented UART, SPI, I2C and CAN bus drivers and verified them with an oscilloscope and logic analyzer.
- Used FreeRTOS tasks, queues and semaphores to process sensor data in real time.
- Followed MISRA C guidelines and fixed 60+ static analysis warnings.
- Debugged hard faults using GDB and JTAG.

### Graduate Engineer Trainee, Sensora Devices

Jul 2022 - May 2023

- Developed Embedded C code for temperature and pressure sensor modules.
- Wrote Python scripts to automate test cases and generate test reports.

## Projects

- **Smart Home Hub:** C++ firmware on ESP32 that reads sensors and reports data over MQTT.
- **Mini RTOS Scheduler:** A round-robin task scheduler in C for a Cortex-M board, built to learn context switching.